

Figure 5: FSInt name clashes

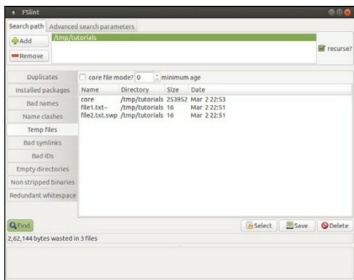


Figure 6: FSInt temp files

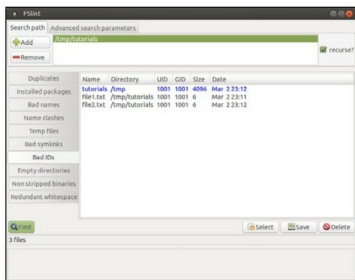


Figure 7: FSInt bad IDs

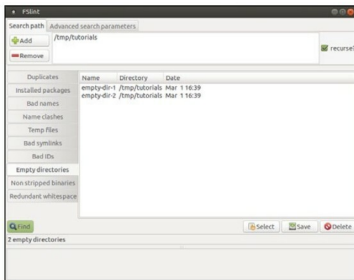


Figure 8: FSInt empty directories

Delete button: As its name suggests, this button is used to delete selected items. I would advise you to perform this operation very carefully, because it deletes files permanently from the system. Before deleting, a confirmation window will appear each time unless the *Ask me this in the future?* check box is unchecked.

Merge button: This operation merges all the files within a group into one physical file using links. To do this, either a hard link or a symbolic link will be used depending on the location of the file. Let us suppose you have two duplicate files, namely, *file1.txt* and *file2.txt*; then the merge operation will delete one of the files and create a link (using the *ln* command) to the other file. If one or more duplicate files are present on the same file system, then the merge option will create a hard link; otherwise, it will create a symbolic link.

Find button: The *Find* button instructs FSInt to perform the selected actions. After completion, it displays the results in the window.

Now that we are familiar with FSInt's GUI interface, let us get some insight into it.

Duplicate files

One of the very common forms of file system lint is duplicate files. We often copy music files, documents, images and videos at multiple locations on the file system while taking backups. As duplicates grow, they eat up available disk space. So let us figure out how to find and remove duplicate files with just a few clicks of the mouse.

The *Duplicates* tab on the left hand side of the screen is the default tab selected at FSInt startup. Just choose the appropriate directory using the *Add* button and then click on the *Find* button. FSInt will show the summary of the duplicate files in the *results* window. Figure 2 shows the result of the operation.

All duplicate files are grouped together under a gray bar giving information such as the number of files in the group and the number of bytes wasted in duplicate files. The total number of bytes wasted in all the files and groups is shown below the *Find* button.

FSInt uses the following algorithm to find duplicate files:

- 1) First, it scans the file system and filters out the files of different sizes.